

The empirical recurrence for OEIS A263962: recovering a conjecture that is not written down

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Abstract

OEIS A263962 records “Empirical recurrence of order 82”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 82, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 82 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A263962 is “Number of $(n+1)X(4+1)$ arrays of permutations of $0..n*5+4$ with each element having index change $(+,-,+,-)$ 0,0 1,1 or 1,2.”. It has offset 1 and begins

465, 23841, 3801496, 406984645, 51140612201, 6038706224797, 730020995021349, 87444873531787712, ...

The entry states:

Empirical recurrence of order 82 (see link above)

As of the “Last modified” line on the live entry (Nov 10 2015 10:23:34, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 82 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{82} c_i a(n-i),$$

c_1	118	c_{22}	969147169186462184804253	c_{43}	−155804022154200802577
c_2	2127	c_{23}	970231428358621403848596	c_{44}	1882548460953297450194
c_3	−251434	c_{24}	−21853087189323512847514088	c_{45}	−285850169828023917305
c_4	2418222	c_{25}	96780840952925083073577519	c_{46}	−9458814330023695915493
c_5	97287877	c_{26}	121999674230571032464701622	c_{47}	10611463280095735779833
c_6	−1281284386	c_{27}	−1802772104960969825859714805	c_{48}	26858845523733132505156
c_7	−15907119343	c_{28}	2743513216505937329805713142	c_{49}	−4535165365954042549346
c_8	321677121988	c_{29}	9499085879968597715184654668	c_{50}	−5172189320824663462015
c_9	−160843061642	c_{30}	−36106031314992565338844415104	c_{51}	12815396285273892326006
c_{10}	−30899161556508	c_{31}	47881300498791675701796554528	c_{52}	72309172290880790264088
c_{11}	−37928125832132	c_{32}	35863863951559978602886503808	c_{53}	−28760176026887653672253
c_{12}	3872778546393564	c_{33}	−602485831863021293257504283136	c_{54}	−8174888229895461145721
c_{13}	11376859061200478	c_{34}	944990456239565043715611830144	c_{55}	412991900758674169256073
c_{14}	−319358239858596494	c_{35}	559873354531171697749646659072	c_{56}	94618822951706107415260
c_{15}	−894710161778357132	c_{36}	−2042846405002144061780446031872	c_{57}	−31012754072570221191427
c_{16}	15987539408815649436	c_{37}	5639710350355334732794541880320	c_{58}	−5567360223852824317714
c_{17}	−15987781479726888708	c_{38}	−10919566159120997303140701454336	c_{59}	99063598679677544637623
c_{18}	−449462363781438481474	c_{39}	18147323560883260096416644837376	c_{60}	−11288505167305470658792
c_{19}	6055284735394485056038	c_{40}	39936529667185911745985162797056	c_{61}	−3211831956503879622359
c_{20}	−975562694838894497931	c_{41}	−114956946568175207078313684893696	c_{62}	21962963278854676987076
c_{21}	−233997516144576625400834	c_{42}	−182219799022153744081064713322496	c_{63}	67801165499428846487333

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 82, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $82 + 4$ terms removed returns order 82 as well, so $\deg q_{\min} = 82 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $82 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 82 that this sequence satisfies — and that family turns out to have exactly one member.

References

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